5 Contrapositive Proof

A. Use the method of contrapositive proof to prove the following statements.

2. Suppose $n \in \mathbb{Z}$. If n^2 is odd, then n is odd.

Proof. We prove the contrapositive: if n is not odd, then n^2 is not odd. Suppose that n is not odd. Thus n is even, so $n = 2a$ for some integer a. Consequently $n^2 = (2a)^2 = 4a^2 = 2(2a^2)$. Therefore $n^2 = 2b$, where the integer $b = 2a^2$. Consequently, n^2 is even by definition of an even number. Therefore n^2 is not odd. $\square$

4. Suppose $a, b, c \in \mathbb{Z}$. If a does not divide bc, then a does not divide b.

Proof. We prove the contrapositive: if $a \mid b$, then $a \mid bc$. Suppose $a \mid b$. Then $b = ak$ for some $k \in \mathbb{Z}$. Multiplying both sides by c gives $bc = a(kc)$. Since $kc \in \mathbb{Z}$, it follows that $a \mid bc$. $\square$